

CHICAGO DUPAGE, IL, USA

WMO: 725305

Lat: 41.914N

Lon: 88.246W

Elev: 754

StdP: 14.30

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94892

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.8	(c) 1.2	(d) -11.4	(e) 3.1	(f) -2.6	(g) -7.4	(h) 3.8	(i) 2.4	(j) 27.3	(k) 25.6	(l) 24.8	(m) 26.1	(n) 9.5	(o) 270	(p) 0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.7	(c) 90.3	(d) 74.7	(e) 87.9	(f) 73.5	(g) 85.1	(h) 71.8	(i) 77.8	(j) 86.8	(k) 75.9	(l) 84.1	(m) 74.3	(n) 82.2	(o) 11.7	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.9	134.6	83.6	73.1	126.3	81.4	71.8	120.9	79.7	41.7	86.8	39.8	84.1	38.2	82.2	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.7	21.3	19.1		-11.5	94.2	6.8	2.8	-16.4	96.2	-20.4	97.8	-24.3	99.4	-29.2	101.4	
				DB												
				WB	-12.1	80.6	6.7	1.6	-16.9	81.8	-20.7	82.7	-24.5	83.6	-29.3	84.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	22.6	26.5	37.3	48.8	59.5	69.2	73.3	71.4	64.9	52.4	39.5	28.4
	DBStd	19.85	12.12	11.33	10.97	9.02	8.61	7.06	5.68	5.19	7.57	8.90	9.59	11.05
	HDD50	3155	849	658	419	127	14	0	0	0	1	79	334	674
	HDD65	6404	1314	1077	859	491	217	38	5	8	95	400	764	1136
	CDD50	3017	0	2	26	91	310	575	722	665	449	154	20	3
	CDD65	793	0	0	2	6	48	163	262	208	93	11	0	0
	CDH74	7261	0	0	14	80	492	1573	2482	1695	823	102	0	0
	CDH80	2274	0	0	1	12	131	528	852	496	236	18	0	0
(14) Wind	WSAvg	9.2	10.6	10.6	10.5	11.1	9.5	8.2	7.3	6.7	7.4	8.9	9.9	9.8
(15) Precipitation	PrecAvg	36.1	1.9	1.7	2.2	3.5	4.2	4.3	3.8	4.1	3.0	3.0	2.6	1.9
	PrecMax	44.3	4.0	5.7	5.0	6.9	7.2	9.9	6.8	9.2	12.7	7.8	5.5	4.7
	PrecMin	23.9	0.5	0.4	0.7	1.2	0.6	1.0	1.6	1.9	0.4	0.7	0.4	0.6
	PrecStd	4.9	1.0	1.1	1.1	1.5	1.7	2.1	1.5	1.7	2.4	1.9	1.5	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	61.5	75.7	81.6	88.3	92.4	95.0	91.2	90.2	82.6	69.3	60.0
		MCWB	51.2	53.2	62.6	64.8	70.2	75.3	77.4	75.0	73.2	69.1	57.8	56.7
	2%	DB	48.2	54.0	66.4	75.3	83.6	89.6	90.5	88.2	86.1	76.6	63.2	53.7
		MCWB	45.1	48.2	56.3	61.1	67.9	73.1	75.9	74.3	71.7	63.7	55.7	49.4
	5%	DB	42.4	46.6	60.9	70.5	80.0	86.1	87.8	85.1	82.1	72.1	59.1	47.8
		MCWB	39.1	41.7	52.9	58.1	65.8	71.8	74.4	72.4	68.9	62.0	53.4	44.7
	10%	DB	38.0	41.3	54.8	65.3	75.1	82.4	84.5	82.3	79.1	67.6	54.6	43.2
		MCWB	35.6	37.4	47.6	55.0	64.1	70.2	72.3	70.7	67.3	60.2	49.1	39.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.1	55.5	63.7	67.6	74.1	79.5	80.3	78.9	76.1	71.4	61.4	57.3
		MCDB	55.1	59.6	73.7	78.7	84.2	89.4	89.8	88.4	86.3	80.4	65.2	59.4
	2%	WB	45.6	48.5	58.9	63.2	70.7	76.1	78.2	76.6	73.5	67.2	57.7	50.4
		MCDB	47.6	53.8	65.1	72.5	80.1	84.8	87.2	84.8	82.1	72.3	61.9	52.2
	5%	WB	39.3	42.3	54.1	60.0	68.2	73.9	76.4	74.7	71.3	64.0	53.7	44.6
		MCDB	41.9	46.8	59.5	68.3	76.5	83.1	84.9	81.9	78.9	69.7	57.5	47.0
	10%	WB	35.8	37.8	47.9	56.3	65.6	71.9	74.7	72.9	69.2	60.7	49.8	39.9
		MCDB	38.1	41.0	54.5	64.6	73.2	80.5	82.4	79.7	76.2	66.7	54.6	43.1
(35) Mean Daily Temperature Range	MDBR		14.3	15.6	18.0	20.8	21.1	20.3	19.7	19.2	21.1	19.4	16.4	13.7
	5% DB	MCDBR	17.8	22.0	25.1	28.8	25.8	22.9	21.3	21.5	24.2	25.1	22.1	17.3
		MCWBR	15.2	17.0	17.0	17.5	13.7	11.3	10.3	10.5	11.8	14.9	16.5	14.9
	5% WB	MCDBR	17.1	20.5	22.6	25.2	22.5	20.1	19.2	18.4	20.5	20.4	19.5	16.3
		MCWBR	15.0	16.6	16.7	17.3	13.5	11.4	11.1	10.7	11.1	14.7	16.2	14.9
(40) Clear-Sky Solar Irradiance	taub		0.293	0.310	0.343	0.397	0.423	0.433	0.441	0.433	0.406	0.353	0.310	0.293
	taud		2.417	2.377	2.384	2.250	2.231	2.242	2.217	2.236	2.286	2.432	2.502	2.446
	Ebn at Noon		272	284	286	277	271	268	264	262	261	265	264	262
	Edh at Noon		25	30	34	42	44	43	44	42	37	29	23	22
(44) All-Sky Solar Radiation	RadAvg		515	802	1140	1471	1684	1921	1929	1689	1404	920	589	430
	RadStd		59	102	93	120	134	134	101	99	120	104	55	45

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+158	N/A
(47) Regional (54 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page